

Bradley Machado

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Data Center Energy & Critical Infrastructure — Utility Rate Strategy · Interconnection · Owner-Side Delivery

SUMMARY

Infrastructure and energy professional with six years operating and planning federal critical facilities, including continuity-of-operations communications and data processing sites. Utility rate and tariff analysis, behind-the-meter storage economics, and Contracting Officer's Representative authority over a privatized utility portfolio. Chaired R&D for Department of the Army capital investment planning at Senior Executive Service and General Officer level. Completing a Master of Architecture in high-performance building design and analysis — energy modeling, lifecycle cost, and the capital decisions that set operating expense.

EXPERIENCE

Research Assistant — Building Energy Analytics | Responsive Architecture Lab, Illinois School of Architecture *Spring 2026 – Present*

- Characterizing the U.S. single-family detached building stock for an HVAC OEM client: harmonizing ~84M property records against the ResStock simulation dataset (~500K modeled homes, National Laboratory of the Rockies, formerly NREL) to transfer peak heating and cooling load estimates onto each record.
- Built in Databricks / PySpark on serverless compute — schema harmonization, categorical crosswalks, and validation at national scale.

Research Planner — Infrastructure & Resilience | U.S. Army ERDC–CERL *2021 – 2025*

- R&D chair for Department of the Army long-term capital investment planning and liaison to IMCOM Headquarters for the Installation Energy and Water Plan and Infrastructure Portfolio Review — identified and evaluated energy solutions, presenting investment options to Senior Executive Service and General Officer decision-makers.
- Secured \$2.075M in competitive funding across two studies, self-funding the position; led cross-agency teams including U.S. Geological Survey and Department of Energy personnel on installation and regional energy and water resilience assessments across North America, Europe, and the Pacific.

Utilities Manager | U.S. Army Garrison – Presidio of Monterey *2019 – 2021*

- Owner's representative for a garrison-wide utility system under master-metered, privatized distribution — a DoD data center, a fixed satellite communications (SATCOM) ground station, public safety and detention facilities, academic and administrative buildings, transit maintenance, and 100+ housing units. Accountable for \$10M in annual utility commodities, \$2M in reimbursables, and \$5M in active construction across mixed load profiles.
- Identified an uncoordinated load doubling at the SATCOM facility during construction planning that had never been raised with the serving utility; as utility subject-matter expert and PG&E point of contact, initiated and drove the distribution backbone upgrade required to support it.
- Cleared FAA obstruction review for pole heights adjacent to an active airfield and NEPA consultation for a protected vernal pool crossing, compressing a standard 3–4 year utility planning cycle to under twelve months and delivering the electrical backbone and a critical gas line with no loss of site uptime.
- Contracting Officer's Representative for 11 privatized utility contracts and the Installation Energy and Water Plan contract; administered Resource Efficiency Management and Utility Energy Service Contract vehicles — task orders, modifications, quality assurance, payment approval, and issue resolution.
- Built a 15-minute interval rate engine evaluating 38 PG&E commercial schedules — legacy A/E and successor B series, Peak Day Pricing variants, and all voltage tiers — against ~102,000 intervals. Scoped to size behind-the-meter storage against the 1 MW rate-class threshold; extended into the portfolio tariff selection and rate migration method.
- Primary point of contact with PG&E for the installation — outage and maintenance coordination, service changes, and billing disputes. Rebuilt utility invoicing and cost tracking end to end; recovered \$200K in lost reimbursables and brought annual cost projections to within 1% of actual.
- Sole 24/7 emergency contact for utility incidents — directed response to sewer, water, gas, and electric failures, including a dormitory evacuation on a struck gas main, and verified backup generation ahead of PG&E outage and brownout events. Held the Public Works seat in the Emergency Operations Center, briefed garrison command, and established response procedures that reduced downtime at critical facilities.
- Coordinated a joint Air Force, Navy, Army, California National Guard, and PG&E project — regional grid upgrade, new generation, solar and battery storage, and a redundant feed. Acting Energy and Utilities Branch Chief, DPW (2020); chaired the Garrison Energy and Water Steering Committee.

Health, Safety & Environmental Manager (Contract) | The Whiting-Turner Contracting Company *2018 – 2019*

- Managed EHS medical operations on the \$3B joint Google and NASA Bayview project; built the incident reporting system used to identify safety hotspots. Streamlined orientation and access, saving 20–30 labor hours daily.

Critical Care Flight Paramedic | U.S. Army — 3rd Infantry Division *2010 – 2014*

- Aeromedical evacuation and critical care; Afghanistan, 2012–2013. Scene medical authority coordinating multinational, federal, state, and civilian agencies.

EDUCATION

Master of Architecture — High-Performance Building Design & Analysis | University of Illinois Urbana-Champaign | Expected 2027

B.S., Society & Environment | University of California, Berkeley | 2018

CLEARANCE & RECOGNITION

- Secret security clearance previously held; eligibility subject to reinstatement verification. OGE-450 confidential financial disclosure filer.
- Department of the Army — Quality Step Increase (off-cycle merit pay increase for exemplary performance) · Civilian Service Commendation Medal (2021) · Civilian Service Achievement Medals (2020, 2019).

PRESENTATIONS & PUBLICATIONS

- Presentations: Leveraging Privatized Utilities for Installation Resilience — DoD Utilities Conference, San Antonio (2019), invited · Ukraine Infrastructure Impacts — Military Operations Research Symposium (2024).
- Publications: REFoRM Policy Analysis — Office of the Assistant Secretary of the Army (2024) · European Regional Water Scarcity Assessment (2024) · Water System Modeling for Increased Resilience — USACE Report (2021).

COMPUTATIONAL METHODS — ENERGY ANALYSIS

- Building energy simulation — EnergyPlus, OpenStudio, Ladybug/Honeybee: envelope, load, and environmental performance modeling from parametric geometry. Multi-objective optimization — Wallacei and Galapagos evolutionary solvers in Grasshopper; Excel Solver for constrained cost optimization.
- Rate and load analysis — Excel (Solver), MATLAB. Data engineering — Python, SQL, Databricks/PySpark. Modeling — Rhino, Grasshopper, Dynamo, Revit, AutoCAD.